PRODUCT(IP/IC) QUALIFICAITON PROCESS

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Quality management Div.

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RS-COP-02 Revision history

Rev. no	Rev. date	Simple description of changed contents	Author	Post
1.0	2025.06.11	Newly established.	Mark.Kim	QM

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1. Purpose and scope

The purpose of this document is to define the standardized qualification process for semiconductor Intellectual Property (IP) and Integrated Circuits (ICs) to ensure that all developed or acquired components meet the required quality, reliability, and performance standards before mass production or customer delivery. This process aims to minimize risk, improve product robustness, and ensure compatibility with manufacturing and end-user environments.

This document applies to all semiconductor IP and IC components developed in-house or sourced from third-party vendors that are intended for integration into SoCs or standalone semiconductor products. The scope includes digital, analog, mixed-signal, and high-speed interface IPs, as well as full ICs across various technology nodes. The qualification process covers design verification, physical implementation checks, silicon validation, reliability testing, and documentation review, and is applicable from early development stages through to production release.

2. Apply scope

This procedure is applied to all of products designed from Ramschip,Inc.

3. Definition of terms

3.1 Qualification

The qualification Refer to the process of verifying that a semiconductor product meets all the necessary requirements and specifications for its intended use. This involves rigorous testing and analysis to ensure that the design functions correctly, performs within specified parameters and meet reliability standards. Qualification typically includes testing for functionality, performance, electrical characteristics, and reliability under various operation conditions. It aims to validate the design's integrity and suitability for mass production, ensuring that the final product meets customer expectations and industry standard.

3.2 Customer satisfaction

A state in which customer demands are being met for tangible and intangible products purchased by customers.

3.3 Silicon proven

Silicon Proven refers to a state in which a semiconductor Intellectual Property (IP) or design has been successfully fabricated, tested, and validated on actual silicon (a physical chip), demonstrating that it meets all functional, performance, and reliability requirements under real-world conditions. This term indicates that the design has moved beyond simulation and pre-silicon validation, and has been confirmed to work correctly through silicon implementation and testing.

3.4 CTS (Compliance Test Specification) Test

Compliance Test Specification refers to a formal document that defines the test procedures, conditions, and criteria used to verify whether a product, component, or system meets the requirements of a specific standard or protocol. This specification ensures that different implementations of a technology remain interoperable, functionally correct, and conform to agreed-upon guidelines set by industry standard organizations.

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4. Responsibility and authority

4.1 CEO

1) Ensure company-wide commitment to quality and reliability

Promote a corporate culture that prioritizes project quality and long-term reliability

Ensure all departments (Design, Quality, Manufacturing, etc.) align with quality objectives.

2) Approve and allocate resources for qualification activities

Ensure sufficient budget, personnel, and infrastructure (e.g. lab equipment, test time) are allocated to perform comprehensive qualification and reliability test.

3) Define and endorse quality and reliability policies

Approve internal quality management policies that include qualification standards, failure criteria, and risk management procedures.

Support adoption of international standard (e.g. ISO9001, IATF16949, ACE-Q100) across the organization.

4) Make business-level decisions based on qualification outcomes

Make go/no-go decisions for product release based on qualification results and associated risks.

Approve escalation plans or mitigation strategies in case of qualification failure.

5) Represent the company in customer and regulatory matters

Serve as the final point of contact for critical customer communications regarding product quality and reliability.

Take responsibility for regulatory compliance and customer audits if escalated.

6) Foster cross-functional accountability

Ensure that all departments collaborate effectively during the qualification process.

Hold executive staff accountable for fulfilling their roles in design, process, quality, and production readiness.

4.2 Design department (Analog solution, Digital solution, Smart solution)

1) Design compliance with project specification

Ensure that circuit and layout design meet the electrical and physical performance requirements as defined by customer and internal product specification

2) Design for reliability

Incorporate considerations for reliability factors such as high temperature, humidity, and electrical stress in to the design.

Include protection against ESD, Latch-up, aging and other reliability risks

3) Provide data for qualification support

Deliver necessary design data for qualification, including corner simulations and worst-case analysis results.

Design test chips or test vehicles required for reliability evaluations.

4) Support failure analysis

Provide technical support form the design perspective when failures occur during qualification.

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Propose corrective action or design improvements as needed.

5) Reflect qualification results in design

Review and implement necessary design changes based on qualification outcome (e.g. margin enhancement, rule updates)

6) Documentation and design history management

Prepare and maintain reliability-related design documents such as design guidelines, DRC/LVS report, and design review records.

4.3 Quality management department

1) Establish and manage qualification plans

Define the qualification strategy, including applicable standard (e.g. JEDEC, AEC Q100) test conditions, sample size and criteria.

Coordinate with design, process and reliability teams to ensure alignment with customers and internal requirements.

2) Monitor and control the qualification process

Ensure that qualification tests are executed properly according to the defined plan and schedule.

Track progress, collect test data, and maintain traceability throughout the qualification cycle.

3) Interface with customers on quality requirements.

Communicate with customers regarding qualification expectations, report formats, schedules, and results.

Address any customer feedback or concerns related to qualification and reliability.

4) Review and approve qualification results.

Analyze test data and determine whether the results meet acceptance criteria.

Provide formal approval (or rejection) of the qualification outcome, including release to production if successful.

5) Drive corrective and preventive actions

When failures occur, lead root cause analysis and corrective action processes across departments.

Ensure that lessons learned are documented and integrated into design, process or test improvements.

6) Maintain quality and reliability documentation

Create and manage qualification reports, failure analysis summaries and control plan.

Ensure documentation complies with internal (Quality management system) and customer audits.

7) Ensure compliance with industry standards

Enforce adherence to relevant standards such as ISO9001, IATF 16949, JEDEC, and AEC-Q series.

Support internal and external audits related to qualification and reliability processes.

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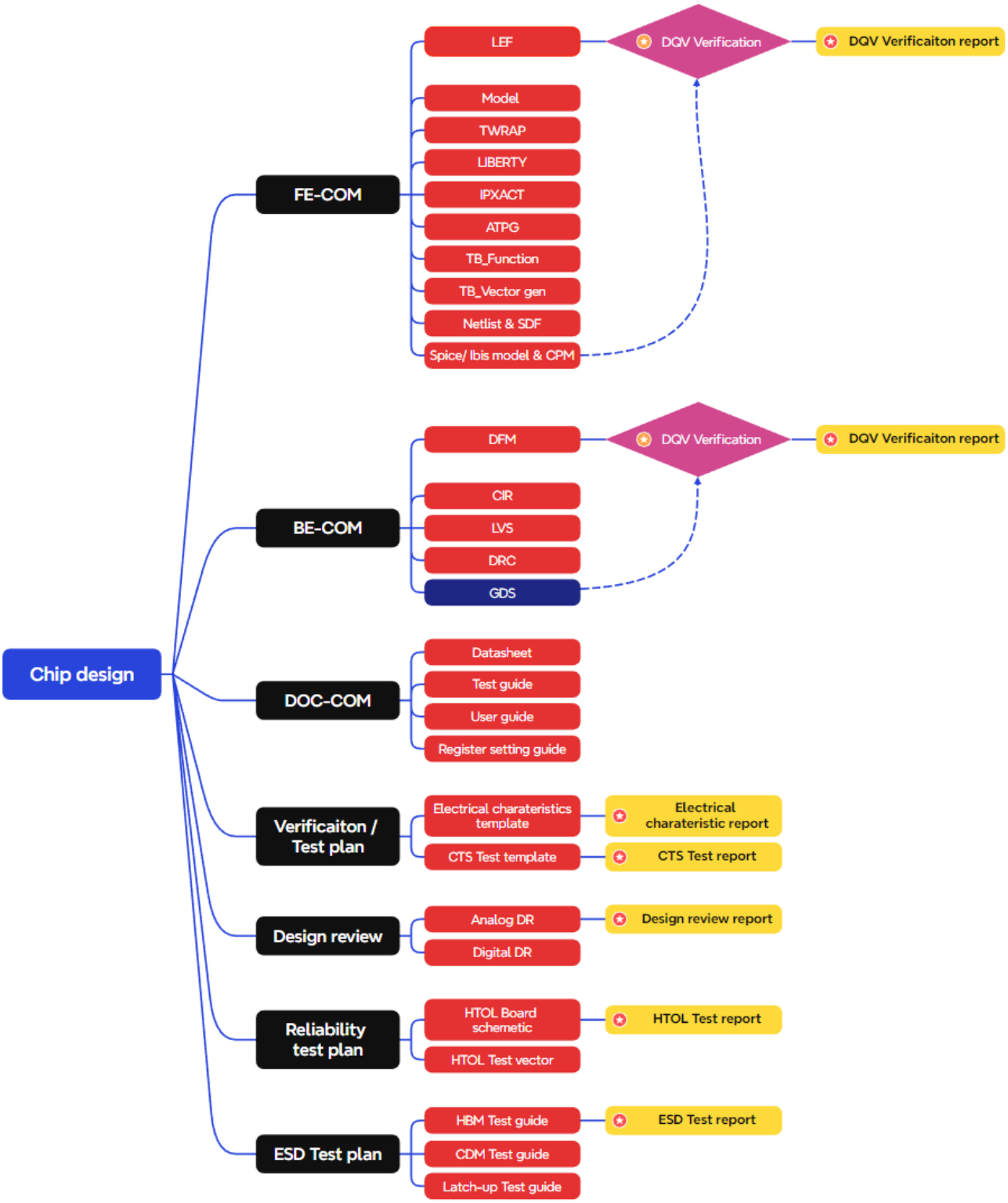
5. Process mapping



6. Work flow description

Phase #1 (DQV Verification)

1) Once the design is completed and the database (DB) is ready, the design department requests DQV verification from the QA department.
2) The QA department conducts the DQV verification on the requested DB and then performs a consistency evaluation of the DB.

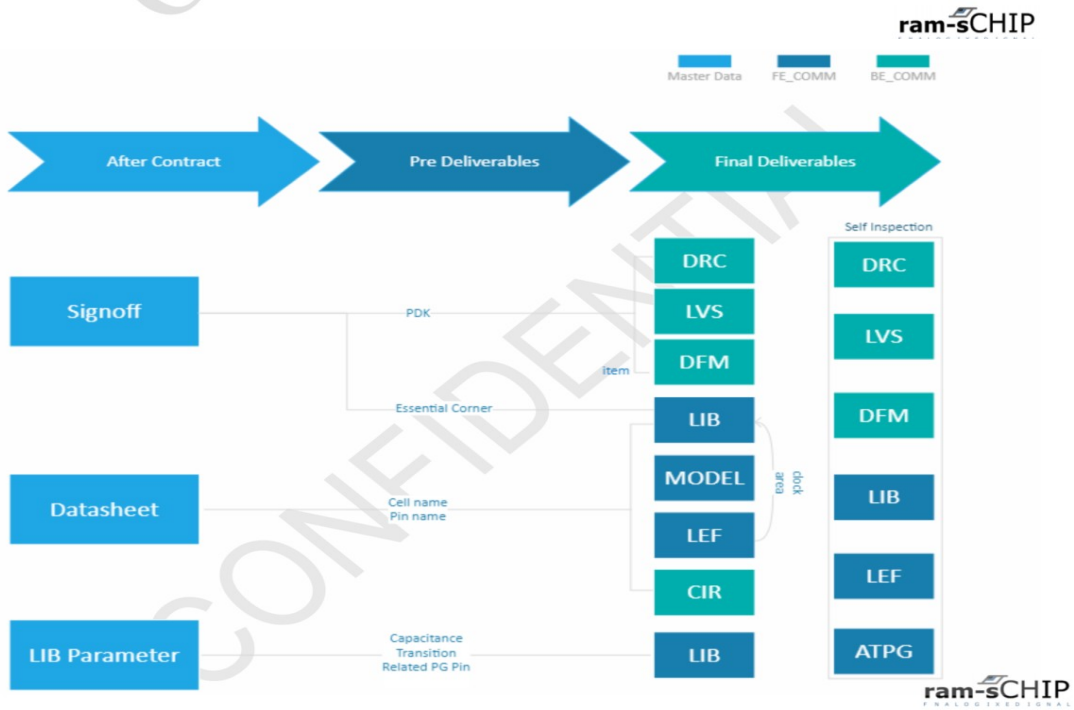
NOTE:

DQV (Design Quality Verification) is an in-house script tool developed by Ramschip to evaluate the consistency of the database (DB). The tool checks pin name matching between various

DQV Rule Decks

Total: 24 rules — **F: FE_COMMON, B: BE_COMMON D: DOC_COMMON**

Rule #	F	B	D	Description	Verification method
1	V	V	V	File name and presence check (Auto load only)	Internal naming convention vs All files in deliverables
2	V	V	V	Cell name and presence check	Internal naming convention vs All files in deliverables
3	V	V	V	Pin name check	CSV vs (MODEL, LEF, LIBERTY, POST_SIM_NET, CIR)
4	V	V	V	Pin direction check	CSV vs (MODEL, LEF, LIBERTY, POST_SIM_NET, CIR)
5	V	V	V	Pin bit width check	CSV vs (MODEL, LEF, LIBERTY, POST_SIM_NET, CIR)
6	V			Cell area check	LEF vs LIBERTY
7	V			LEF Cell Consistency	Comparing within LEF files
8	V		V	P/G Pin Check	Comparing within LIB files (P/G Pin vs Voltage Map)
9	V			Liberty description (PVT) check	Comparing within LIB files (Library vs default_operating_conditions)
10	V			Essential Corner check	Signoff vs LIBERTY
11	V			Clock check	Comparing the clock pins (in the LEF file vs in the LIB files)
12	V			No timing arc	Comparing within LIB files
13	V			Timing arc with numbers	Comparing within LIB files
14	V			Intra-Liberty Related P/G Pin check	Voltage map vs LIB files
15	V			Related Pin	Comparing within LIB files
16	V			Max Capacitance check	Specified Value vs LIB files
17				Self-Capacitance check	Specified Value vs LIB files
18	V			Max Transition	Specified Value vs LIB files
19		V		PDK Version check	Signoff VS (LVS, DRC, DFM)
20	V			Test Coverage	ATPG Files
21		V	V	DRC Signoff Check	Signoff VS DRC (sourceme. *)
22		V	V	DFM Item check	Signoff VS DFM
23		V	V	LVS, DFM Error Log check	Within LVS, DFM Report
24		V	V	Bump Size Check	Customer Requirement (Orderform) VS DRC (sourceme.*)



deliverables and verifies simulation results.

Manual Link: [DQV User Manual V1.01.pdf]

Phase #2 (Electrical characteristics evaluation)

1) After the test chip is fabricated (fab-out) based on the designed database, it proceeds to package assembly.
2) QA engineer proceeds the electrical characteristics (ex/ standby current, dynamic current, PLL,

Silicon Proven check list

1.1. General information

IP	RS_DCPHY6310_LN04LPP_M4S4	PDK Ver.	S00	Evaluation engineer	DS - Ian/ Brett	Final measurement date	
Sample package	144FCBGA	Corner sample	SS	QA Check engineer	QM - Ronnie	Final checking date	

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1.2. Silicon proven items & result

Clasification	Test items	Probe point	Vector	Symbol	Power Condition	Judgement criteria min	typ.	max.	unit	Measurement 2st Build Sample SS #1 -40 ℃	RT	125℃	2st Build Sample SS #2 -40 ℃	RT	125℃	2st Build Sample SS #11 -40 ℃	RT	125℃	CT MIN	CT MAX	RT MIN	RT MAX	HT MIN	HT MAX	Judgement (Pass/Fail) Sample #1	Sample #2	Sample #3
Scan Test	SCAN_TEST	M_SCAN_OUT_C/0/1/2/3_TM[X:0] S_SCAN_OUT_C/0/1/2/3_TM[X:0]	-	ALL and SCAN	Condition #1 MS_VDD0P85 (0.8925V)	Pass/Fail			-	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
					Condition #2 MS_VDD0P85 (0.7125V)	Pass/Fail			-	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
IDS Current	IDS current without power gating	MS_VDD0P85	ids.v	IDS_MS_VDD0P85	MS_VDD0P85 (0.85V) M_VDD12 (1.2V)	-		60	mA	0.9	1.8	9.7	0.8	1.4	7.8	0.8	1.6	10.1	0.80	0.90	1.40	1.80	7.80	10.10	On going	On going	
		M_VDD12		IDS_M_VDD12		-		10		0	0.002	0.002	0.001	0.001	0.002	0.001	0.001	0.002	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	
	IDS current with power gating	MS_VDD0P85		IDS_MS_VDD0P85		-		40		0.3	0.9	4.9	0.4	0.9	4.1	0.4	0.9	5.2	0.40	0.50	0.90	0.90	4.10	5.20	On going	On going	
		M_VDD12		IDS_M_VDD12		-		10		0	0.002	0.002	0.001	0.001	0.002	0	0.001	0.002	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	
	Slave pad leakage	Slave DP and DN pads		ILEAK		-		100	uA	0.1	0.1	0.35	0.1	0.13	0.42	0.11	0.11	0.37	0.10	0.11	0.10	0.13	0.35	0.42	On going	On going	
	IDS current without power gating	MS_VDD0P85	ids.v	IDS_MS_VDD0P85	MS_VDD0P85 (0.7125V) M_VDD12 (1.14V)	-		60	mA										0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
		M_VDD12		IDS_M_VDD12		-		10											0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
	IDS current with power gating	MS_VDD0P85		IDS_MS_VDD0P85		-		40											0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
		M_VDD12		IDS_M_VDD12		-		10											0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
	Slave pad leakage	Slave DP and DN pads		ILEAK		-		100	uA										0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
	IDS current without power gating	MS_VDD0P85	ids.v	IDS_MS_VDD0P85	MS_VDD0P85 (0.8925V) M_VDD12 (1.26V)	-		60	mA										0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
		M_VDD12		IDS_M_VDD12		-		10											0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
	IDS current with power gating	MS_VDD0P85		IDS_MS_VDD0P85		-		40											0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
		M_VDD12		IDS_M_VDD12		-		10											0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
	Slave pad leakage	Slave DP and DN pads		ILEAK		-		100	uA										0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
DC-PHY Master Core IP Unidirectional LP Mode Test	Master clock lane LP-TX function	M_DPCLK M_DNCLK M_TEST_OUT_C[1:0]	dphy_master_clock_lptx	LPTX_MC_DPHY	Condition #1 MS_VDD0P85 (0.8925V)	Pass/Fail			-	PASS	PASS	PASS		PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
				LP_VOH		1.08	1.2	1.32	V	1.24	1.24	1.24	1.21	1.21	1.24	1.23	1.24	1.24	1.21	1.24	1.21	1.24	1.24	1.24	On going	On going	On going
				LP_VOL		-0.05	0	0.05		0.001	0.001	0.001	0	0.001	0.001	0	0.001	0.001	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
				LPTX_MC_DPHY	Condition #2 MS_VDD0P85 (0.7125V)	Pass/Fail			-	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
				LP_VOH		1.08	1.2	1.32	V	1.12	1.12	1.12	1.09	1.083	1.12	1.11	1.12	1.12	1.09	1.12	1.08	1.12	1.12	1.12	On going	On going	On going
				LP_VOL		-0.05	0	0.05		0.001	0.001	0.001	0	0.001	0	0	0.001	0.001	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
	Master data lane LP-TX function	M_DPDATA0/1/2/3 M_DNDATA0/1/2/3 M_TEST_OUT_0/1/2/3[1:0]	dphy_master_data_lptx	LPTX_MD_DPHY	Condition #1 MS_VDD0P85 (0.8925V)	Pass/Fail			-	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
				LP_VOH		1.08	1.2	1.32	V	1.24	1.24	1.24	1.21	1.21	1.24	1.23	1.24	1.24	1.21	1.24	1.21	1.24	1.24	1.24	On going	On going	On going
				LP_VOL		-0.05	0	0.05		0.001	0.001	0.001	0	0.001	0.001	0	0.001	0.001	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
				LPTX_MD_DPHY	Condition #2 MS_VDD0P85 (0.7125V)	Pass/Fail			-	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
				LP_VOH		1.08	1.2	1.32	V	1.12	1.12	1.12	1.09	1.084	1.12	1.11	1.12	1.12	1.09	1.12	1.08	1.12	1.12	1.12	On going	On going	On going
				LP_VOL		-0.05	0	0.05		0.001	0.001	0.001	0	0.001	0	0	0.001	0.001	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
	Master data lane LP-TX function(C-PHY)	M_DNDATA0_A0 M_DPDATA0_B0 M_DNDATA1_C0 M_DPDATA1_A1 M_DNCLK_B1 M_DPCLK_C1	cphy_master_data_lptx	LPTX_MD_CPHY	Condition #1 MS_VDD0P85 (0.8925V)	Pass/Fail			-	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
				LP_VOH		1.08	1.2	1.32	V	1.24	1.24	1.24	1.21	1.21	1.24	1.23	1.24	1.24	1.21	1.24	1.21	1.24	1.24	1.24	On going	On going	On going
				LP_VOL		-0.05	0	0.05		0	0	0.001	0	0	0.001	0	0	0.001	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
				LPTX_MD_CPHY	Condition #2 MS_VDD0P85 (0.7125V)	Pass/Fail			-	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
				LP_VOH		1.08	1.2	1.32	V	1.12	1.12	1.12	1.09	1.090	1.12	1.11	1.12	1.12	1.09	1.12	1.09	1.12	1.12	1.12	On going	On going	On going
				LP_VOL		-0.05	0	0.05		0	0	0	0	0	0	0	0	0	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
DC-PHY Master Core IP Bidirectional LP Mode Test	Mater data lane LP-RX function(D-PHY)	M_TEST_OUT_0[0]	dphy_master_data_lprx	MD_LPRX_DPHY	Condition #1 MS_VDD0P85 (0.8925V) M_VDD12 (1.26V)	Pass/Fail			-	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
	Mater data lane LP-RX function(C-PHY)	M_TEST_OUT_0[0]	cphy_master_data_lprx	MD_LPRX_CPHY						PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
	D-PHY master data lane LP-CD function	M_TEST_OUT_0[1]	dphy_master_data_lpcd	MD_LPCD_DPHY						PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
	C-PHY master data lane LP-CD function	M_TEST_OUT_0[1]	cphy_master_data_lpcd	MD_LPCD_CPHY						PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
	Mater data lane LP-RX function(D-PHY)	M_TEST_OUT_0[0]	dphy_master_data_lprx	MD_LPRX_DPHY	Condition #2 MS_VDD0P85 (0.7125V) M_VDD12 (1.14V)	Pass/Fail			-	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
	Mater data lane LP-RX function(C-PHY)	M_TEST_OUT_0[0]	cphy_master_data_lprx	MD_LPRX_CPHY						PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
	D-PHY master data lane LP-CD function	M_TEST_OUT_0[1]	dphy_master_data_lpcd	MD_LPCD_DPHY						PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
	C-PHY master data lane LP-CD function	M_TEST_OUT_0[1]	cphy_master_data_lpcd	MD_LPCD_CPHY						PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
DC-PHY Slave Core IP Uni-direction LP Mode Test	Slave clock lane LP-RX function(D-PHY)	S_TEST_OUT_C[1:0]	dphy_slave_clock_lprx	SC_LPRX_DPHY	Condition #1 MS_VDD0P85 (0.8925V)	Pass/Fail			-	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
	Slave data lane LP-RX function(D-PHY)	S_TEST_OUT_0/1/2/3[1:0]	dphy_slave_data_lprx	SD_LPRX_DPHY						PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
	Slave data lane LP-RX function(C-PHY)	S_TEST_OUT_0/1/2[1:0]	cphy_slave_data_lprx	SD_LPRX_CPHY						PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
	Slave clock lane LP-RX function(D-PHY)	S_TEST_OUT_C[1:0]	dphy_slave_clock_lprx	SC_LPRX_DPHY	Condition #2 MS_VDD0P85 (0.7125V)	Pass/Fail			-	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
	Slave data lane LP-RX function(D-PHY)	S_TEST_OUT_0/1/2/3[1:0]	dphy_slave_data_lprx	SD_LPRX_DPHY						PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
	Slave data lane LP-RX function(C-PHY)	S_TEST_OUT_0/1/2[1:0]	cphy_slave_data_lprx	SD_LPRX_CPHY						PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	0.00	0.00	0.00	0.00	0.00	0.00	On going	On going	On going
PLL Frequency	PLL 6.5GHz (DPHY)	PLL_TEST_DIV	dphy_ext(int)_lb_6500M.v	PLL_FREQ_DPHY_6p5G	Condition #1 MS_VDD0P85 (0.8925V)	#####	#####	######	MHz	101.280	101.360	101.309	101.280	#####	#####	#####	######	######	101.28	101.38	101.30	101.36	101.19	101.38	On going	On going	On going
					Condition #2 MS_VDD0P85 (0.8075V)					101.376	101.299	101.306	101.312	#####	#####	#####	######	######	101.31	101.38	101.30	101.38	101.30	101.37	On going	On going	On going
	PLL 2.5GHz (DPHY)		dphy_ext(int)_lb_2500M.v	PLL_FREQ_DPHY_2p5G	Condition #1 MS_VDD0P85 (0.8925V)	38.770	38.965	39.160		38.944	38.928	38.922	39.072	38.922	39.107	39.072	39.060	39.066	39.07	39.07	38.92	39.06	38.92	39.11	On going	On going	On going
					Condition #2 MS_VDD0P85 (0.7125V)					38.944	38.915	38.930	38.994	38.954	39.062	39.072	39.056	39.069	38.94	39.07	38.92	39.06	38.93	39.07	On going	On going	On going

Function loop-back test, etc.) based on the test plan or test checklist.

3) If the measured results fall outside the acceptance criteria, The QA engineer notifies the design department, and they conduct a root cause analysis.
4) If the items outside the acceptance criteria can be de-bugged, the engineers should perform the debugging and then request re-qualification from the QA engineer.
5) QA engineer performs re-qualification if the troubleshooting is deemed reasonable.

Phase #3 (compliance Test Suite)

1) Once the electrical characteristics evaluation is complete, QA / Design engineers should perform compliance test (CTS) to ensure that the designed IP meet the required standard.
2) CTS test should be performed based on the items and criteria defined in the CTS standard.
3) The linked CTS report follows Ramschip's official report format.

[CTS REPORT_rs_dcphy6310a_ln04lpp.pdf]

Phase #4 (ESD Test / Device life Test)

1) The QA engineer should prepare samples to enable the execution of the ESD/HTOL test.
2) The QA/Design engineer prepares a test plan outlining how the ESD/HTOL Test will be conducted.

3) The QA engineer provides the ESD/RT test plan to a testing company such as QRT and requests the execution of the test.
   Refer to the follow test guide.
   Once the ESD test is completed by ESD test site, the QA engineer performs post measurement on the zapped samples.
4) After making the final judgment on the test results, the QA engineer prepares the test report.

Phase #4-1 ESD / HTOL Test guide

ESD Model	Test pin group	Ref. GND Pin group	Test level	Zap	Sample size	Judgment
HBM (Human Body Model)	VSS, IO	VDD	Target level (KV)	(+/-) 1	3	0/3
	VDD, IO	VSS, VSS_no_zap		(+/-) 1	3	0/3
	IO	IO		(+/-) 1	3	0/3
Note: 1st step _ Conduct an ESD limit test with a single sample each mode. 2nd step _ Conduct an ESD test with target level for qualification each mode						
CDM (Charged Device Model)	ALL Pin	N/A	Peak current 7A level	(+/-) 1	3	0/3
Note: 1st step _ To determine the for a peak current of 7A, CDM zapping test will be conducted. 2nd step _ Check for damage by measuring the pin I-V curve. 3rd step _ Conduct an CDM test with the found test level for qualification.						
Latch-up	Temperature condition: Room		100mA	Positive 1	3	0/3
				Negative 1	3	0/3
			Vmax *1.5	(+) 1	3	0/3

Reliability	Temperature	Stress voltage	Stress time	Sample size	Judgment
HTOL (High temperature operating life)	125 C	Operating Max V or the stress condition equivalent 10years operating life.	504hr / 1008hr	77	0/77

Phase #4-2 Guideline for judgment criteria

Test items	Input Voltage	0hr measurement	Post measurement	Criteria information

ESD_Latch-up_Guide.pptx

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Standby current (IDDQ)	Typ. V	0hr Current	Post current	**[Commercial IP or product]** De-gradation: x2 ↑ **[Automotive IP or product]** De-gradation: x1.5 ↑
Dynamic current (DIDD)	Min. V Max. V	0hr Current	Post current	**[Commercial IP or product]** De-gradation: 30% ↑ **[Automotive IP or product]** De-gradation: 20% ↑
Scan test	Min. V Max. V	Good die	Pass/fail under input voltage	Pass / Fail
Function test (External loopback test)	Min. V Max. V	Good die	Pass/fail under input voltage	Pass / Fail

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7. Relevant document

Material	Form code	Description
ESD_Latch-up_Guide	RS-MP-06-I-01	-. Explanation of ESD test Theory and test methods.

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